Important Questions – Explained

Module 1: Polar Curves & Curvature

Every solved problem in your notes, grouped by pattern, with the recognition cue and method for each

How This Module's Questions Are Structured

Your notes contain roughly **4 question families**. Once you recognize which family a question belongs to, the method is fixed – only the algebra changes:

1. Find ϕ (angle between radius vector & tangent) for one curve, possibly at a specific θ.
2. Show two curves cut **orthogonally**.
3. Find the **angle between** two curves (general, not necessarily 90°).
4. Find the **pedal equation** of a curve.
5. Find the **radius of curvature** of a curve (Cartesian, polar, parametric, or pedal-equation route).

1 Family 1: Find ϕ for a Single Curve

Recognize this question by: "Find the angle between the radius vector and tangent to the curve $r = f(\theta)$; also find ϕ at $\theta = \dots$"

1. Differentiate $r = f(\theta)$ to get $dr/d\theta$.
2. Form $\cot\phi = \dfrac{1}{r}\dfrac{dr}{d\theta}$ and simplify algebraically until it matches a clean cot(something) or tan(something) form.
3. Solve for ϕ.
4. If a specific θ is given, substitute at the very end.

Worked pattern – $r = a(1 - \cos\theta)$, find ϕ at $\theta = 60°$:

$$\frac{dr}{d\theta} = a\sin\theta \;\Rightarrow\; \cot\phi = \frac{1}{r}\frac{dr}{d\theta} = \frac{a\sin\theta}{a(1-\cos\theta)} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} = \cot\frac{\theta}{2} \;\Rightarrow\; \phi = \frac{\theta}{2}$$

At $\theta = 60° = \pi/3$: $\phi = \pi/6$.

Key trick: Whenever you see $1 - \cos\theta$ or $1 + \cos\theta$ in a denominator, immediately convert using the half-angle identities $1 - \cos\theta = 2\sin^2(\theta/2)$ and $1 + \cos\theta = 2\cos^2(\theta/2)$. The numerator $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$ always cancels one factor cleanly, leaving a simple tan or cot of $\theta/2$. This single trick solves the majority of cardioid-type problems in your notes.

Second worked pattern – $r = a(1 + \cos\theta)$, find ϕ at $\theta = \pi/6$: here the sign flips because of the $+\cos\theta$, giving $\cot\phi = -\tan(\theta/2)$, which needs the identity $-\tan x = \cot(\pi/2 + x)$ to convert cleanly, giving $\phi = \dfrac{\pi}{2} + \dfrac{\theta}{2}$. At $\theta = \pi/6$: $\phi = \dfrac{7\pi}{12}$.

2 Family 2: Show Two Curves Cut Orthogonally

Recognize this question by: "Show that the given pairs of polar curves intersect orthogonally," listing 2 curve equations with a shared parameter.

1. Find ϕ_1 for curve 1 (Family 1 method).
2. Find ϕ_2 for curve 2 the same way.
3. Compute $|\phi_2 - \phi_1|$. If it simplifies to $\pi/2$ **for all** θ (the θ-dependence cancels out), the curves are orthogonal everywhere they meet.

Six pairs from your notes, and the trick each one needs:

(i) $r = a(1 - \cos\theta)$, $r = b(1 + \cos\theta)$: half-angle trick gives $\phi_1 = \theta/2$ and $\phi_2 = \pi/2 + \theta/2$; difference is exactly $\pi/2$ – notice the different constants a, b never even entered the calculation, since $\cot\phi$ only depends on the θ-shape, not the size.

(ii) $r = a(1+\sin\theta)$, $r = b(1-\sin\theta)$: same idea but with $\sin\theta$ half-angle identities ($1 \pm \sin\theta = (\cos\frac{\theta}{2} \pm \sin\frac{\theta}{2})^2$) – messier algebra, same structure.

(iii) $r = \dfrac{a}{1+\cos\theta}$, $r = \dfrac{b}{1-\cos\theta}$: these are conics (parabola-type). Differentiating $r(1+\cos\theta) = a$ implicitly (product rule on the LHS, since a is constant) is required instead of just differentiating $r = f(\theta)$ directly.

(iv) $r^n = a^n \cos n\theta$, $r^n = b^n \sin n\theta$: differentiate implicitly using $nr^{n-1}\dfrac{dr}{d\theta} = \ldots$, giving $\cot\phi_1 = -\tan n\theta$ and $\cot\phi_2 = \cot n\theta$ – the angle works out to $\pi/2$ regardless of n.

(v) $r^n \cos n\theta = a^n$, $r^n \sin n\theta = b^n$: nearly identical to (iv) but the constant sits outside instead of as the base – same differentiation strategy.

(vi) $r = 4\sec^2(\theta/2)$, $r = 9\csc^2(\theta/2)$: first rewrite as $r(1+\cos\theta) = 8$ and $r(1-\cos\theta) = 18$ using $\sec^2, \csc^2$ half-angle identities – then it reduces to case (i)'s method.

Key trick: In every orthogonality proof, the constants (a, b, or the specific numbers like 4 and 9) are a distraction – they always cancel when you form $\cot\phi = \frac{1}{r}\frac{dr}{d\theta}$, because that ratio only cares about the functional shape in θ. If your algebra doesn't show the constant cancelling, you've made an error.

3 Family 3: Angle Between Two Curves (General)

Recognize this question by: "Find the angle between the given polar curves" (no mention of orthogonal/perpendicular).

Same as Family 2, but the final $|\phi_2 - \phi_1|$ will **not** simplify to a constant $\pi/2$ – instead you'll need to either (a) leave the answer in terms of θ, or (b) first solve the two curve equations simultaneously to find the specific point of intersection, then substitute.

Four worked patterns from your notes:

(i) $r = 2\sin\theta$, $r = 2\cos\theta$: both simplify instantly to $\phi_1 = \theta$ and $\phi_2 = \pi/2 + \theta$, giving a constant angle $\pi/2$ – so even a "find the angle" question can turn out orthogonal; don't assume non-orthogonal just because the question doesn't say so.

(ii) $r = a\log\theta$, $r = a/\log\theta$: differentiating gives $\tan\phi_1 = \theta\log\theta$ and $\tan\phi_2 = -\theta\log\theta$. Using the compound-angle formula for $\tan(\phi_1 - \phi_2)$ gives an expression in θ – **then** you must find the intersection point by setting the two curve equations equal: $a\log\theta = a/\log\theta \Rightarrow (\log\theta)^2 = 1 \Rightarrow \theta = e$. Substitute $\theta = e$ into the angle expression to get the final numeric answer $2\tan^{-1} e$.

(iii) $r^2 \sin 2\theta = 4$, $r^2 = 16\sin 2\theta$: implicit differentiation (chain rule on r^2) gives $\cot\phi_1 = -\cot 2\theta$ and $\cot\phi_2 = \cot 2\theta$, so $|\phi_2 - \phi_1| = 4\theta$ symbolically – then solve the two equations together ($4/\sin 2\theta = 16\sin 2\theta \Rightarrow \sin^2 2\theta = 1/4$) to get a specific $\theta = \pi/12$, giving final angle $\pi/3$.

(iv) $r = a(1-\cos\theta)$, $r = 2a\cos\theta$: gives $\phi_1 = \theta/2$ (half-angle trick) and $\phi_2 = \pi/2 + \theta$ (implicit differentiation), difference $= \pi/2 + \theta/2$ symbolically – then solve $a(1-\cos\theta) = 2a\cos\theta \Rightarrow \cos\theta = 1/3$ for the intersection, and substitute back.

Key trick: The moment your simplified $|\phi_2 - \phi_1|$ still contains θ (doesn't cancel to a constant), that's your signal you must go find the actual intersection point next – set the two given curve equations equal to each other and solve for θ.

4 Family 4: Pedal Equation of a Curve

Recognize this question by: "Find the pedal equation of the curve $r = f(\theta)$" or a request to prove a relation like $2ap^2 = r^3$.

1. Differentiate the curve to find $\cot\phi = \dfrac{1}{r}\dfrac{dr}{d\theta}$ (exactly as in Family 1).
2. Substitute into $\dfrac{1}{p^2} = \dfrac{1}{r^2}(1 + \cot^2\phi)$.
3. Simplify, then use the **original curve equation** to eliminate any remaining θ, leaving only r and p.

Six worked patterns from your notes:

(i, ii) $r = a(1 \pm \cos\theta)$ (cardioids): half-angle trick gives $\cot\phi = \mp\tan(\theta/2)$ or $\cot(\theta/2)$; after substituting into the pedal formula and simplifying with $1 \pm \cos\theta = r/a$, you land on the compact result $2ap^2 = r^3$.

(iii) $r^m = a^m(\cos m\theta + \sin m\theta)$: differentiate implicitly ($mr^{m-1}dr/d\theta = a^m m(-\sin m\theta + \cos m\theta)$), then use $\cos^2 + \sin^2 = 1$ to simplify $1 + \cot^2\phi$ into a clean fraction, giving $2a^{2m}p^2 = r^{2(m+1)}$.

(iv) $r = ae^{\theta\cot\alpha}$ (equiangular/logarithmic spiral): differentiating an exponential gives $\cot\phi = \cot\alpha$ directly (a **constant** angle $\phi = \alpha$ – this is the defining property of this spiral, every tangent makes the same angle with the radius vector). Pedal equation becomes simply $p = r\sin\alpha$.

(v) $l/r = 1 + e\cos\theta$ (general conic with eccentricity e): differentiating and simplifying $1 + \cot^2\phi$ using $\cos^2\theta + \sin^2\theta = 1$ leads, after replacing $\cos\theta$ using the original equation, to $\dfrac{1}{p^2} = \dfrac{1}{l^2}\left(e^2 + \dfrac{2l}{r} - 1\right)$ – the standard conic pedal equation (important for orbital mechanics / Kepler-type applications).

(vi) $r^m = a^m\cos m\theta + b^m\sin m\theta$: differentiate, form $1 + \cot^2\phi$, use $\cos^2 m\theta + \sin^2 m\theta = 1$ on the numerator to collapse the cross terms, giving $\dfrac{1}{p^2} = \dfrac{a^{2m} + b^{2m}}{r^{2m+2}}$.

Key trick: No matter how complicated the curve looks, the pedal-equation method is always the same 3 steps. The only real skill being tested is algebraic simplification – especially spotting where $\sin^2 + \cos^2 = 1$ can collapse a messy numerator, and knowing to substitute the original curve equation back in in the last step to kill leftover θ's.

5 Family 5: Radius of Curvature

Recognize this question by: "Find the radius of curvature of the curve..." – then check what form the curve is given in to pick the right formula (see Study Guide, Section 4).

5a. Cartesian-form problems – use $\rho = \dfrac{(1 + y_1^2)^{3/2}}{y_2}$

- $y = a\log(\sec(x/a))$: $y_1 = \tan(x/a)$, $y_2 = \frac{1}{a}\sec^2(x/a) \Rightarrow \rho = a\sec(x/a)$.
- $y^2 = 4ax$ (parabola): implicit differentiation twice; simplifies to $\rho = \dfrac{(y^2 + 4a^2)^{3/2}}{4a^2}$.
- $y = c\cosh(x/c)$ (catenary): uses $\cosh^2 - \sinh^2 = 1$; simplifies beautifully to $\rho = y^2/c$.
- $x^3 + y^3 = 3axy$ (folium of Descartes) at $(3a/2, 3a/2)$: implicit differentiation is messy – differentiate once for y_1, evaluate at the point first (often gives a clean number like $y_1 = -1$), **then** differentiate again for y_2 using the already-simplified y_1 – don't try to get a general y_2 formula before substituting the point.
- $a^2y = x^3 - a^3$ where the curve cuts the x-axis: first find the point (set $y = 0$, solve for x), then proceed as usual.
- $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at $(a/4, a/4)$: implicit differentiation with fractional powers – same "substitute the point early" strategy as the folium.
- $x^2y = a(x^2 + y^2)$ at $(-2a, 2a)$: same strategy again.

Key trick: Whenever a specific point is given (not just "find ρ at any point"), substitute the point's coordinates into y_1 **as soon as you compute it**, before differentiating again for y_2. This keeps the second differentiation dramatically simpler than carrying the full symbolic y_1 through.

5b. Polar-form problems – use $\rho = \dfrac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r\,r_2}$

- $r^n = a^n\cos n\theta$ and $r^n = a^n\sin n\theta$: both give the clean proportionality result $\rho \propto \dfrac{1}{r^{n-1}}$ – a good one to recognize by pattern rather than fully re-deriving under time pressure.

5c. Parametric-form problems – use $\rho = \dfrac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$

- Cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$: after simplifying y_1 using half-angle identities, and squaring ρ's numerator using $1 + \cos\theta$ again, the answer collapses to $\rho = 2^{3/2}a(1 + \cos\theta)^{-1/2}$ or equivalently $4a\cos(\theta/2)$ – the half-angle trick from Family 1 reappears here too.

5d. Pedal-equation-route problems – use $\rho = r\,\dfrac{dr}{dp}$

- $2a/r = 1 + \cos\theta$, show $\rho^2 \propto r^3$: first derive the pedal equation ($p^2 = ar$, using the Family 4 method), then differentiate that pedal relation with respect to p to directly get $\rho = r\,dr/dp \Rightarrow \rho^2 = Kr^3$. This is faster than differentiating the original polar curve twice.

- $2a/r = 1 - \cos\theta$: same approach, sign flips.
- $r = a(1 - \cos\theta)$ (cardioid), prove $\rho^2 \propto r$: pedal equation first ($2ap^2 = r^3$ from Family 4(i)), then differentiate for ρ.
- $\theta = \dfrac{\sqrt{r^2 - a^2}}{a} - \cos^{-1}(a/r)$: here θ is given as a function of r (inverted form) – differentiate to get $d\theta/dr$, take the reciprocal for $\cot\phi$, simplify (the $\sqrt{r^2 - a^2}$ terms partially cancel) to get $p = \sqrt{r^2 - a^2}$, then $\rho = r\,dr/dp = \sqrt{r^2 - a^2}$.

Key trick: If a question explicitly asks you to prove $\rho^2 \propto r^n$ (a proportionality, not a numeric answer), that is a strong signal to find the pedal equation **first** and differentiate that, rather than grinding through the polar radius-of-curvature formula directly – it's almost always faster.

Exam Strategy Summary

1. **Identify the family first** (single-curve ϕ, orthogonality, general angle, pedal equation, or radius of curvature) before writing anything – this tells you which formula box to start from.
2. **For any curve with** $1 \pm \cos\theta$ **or** $1 \pm \sin\theta$, apply half-angle identities immediately – this single move resolves most of Families 1–3.
3. **For "angle between curves" questions**, check whether the θ cancels out of $|\phi_2 - \phi_1|$. If yes, you're done. If no, go find the intersection point by solving the two curve equations together.
4. **For pedal equations**, always finish by substituting the original curve equation back in to eliminate leftover θ – an answer still containing θ is incomplete.
5. **For radius of curvature "prove** $\rho^2 \propto r^n$**" questions**, derive the pedal equation first, then use $\rho = r\,dr/dp$ – it's shorter than direct polar differentiation.
6. **For Cartesian radius-of-curvature at a specific point**, substitute the point into y_1 before computing y_2, not after.

Based on the solved problems in the Module 1 (Differential Calculus – Polar Curves) lecture notes prepared by Purushotham P, Dept. of Mathematics, S.J.C. Institute of Technology, Chickballapur, for 22MATM11.